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The impact of the market-oriented and planning-oriented allocation of medical resources diversity on real estate prices

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ABSTRACT

The impact of public facilities on real estate prices is a crucial consideration. In particular, medical facilities have been identified as a significant factor in the fluctuation of real estate prices. This study not only evaluates the effect of different levels of medical facilities on real estate prices, but also considers medical diversity and its spatial distribution characteristics. In this article, medical diversity refers to various medical service providers, including medical centres, local hospitals, regional hospitals, health centers, clinics, and Pharmacies. The multiscale impact of medical diversity on real estate prices is explored through multi-stage data mining and multiscale geographically weighted regression (MGWR) models. The results show that most medical resources in Taiwan are market-oriented in spatial distribution, which significantly increases real estate prices. In contrast, local hospitals only provided by policy-based medical resources have a more significant impact on low housing prices. This study demonstrates the spatial patterns of medical diversity, which contributes to planning decisions on equitable resource allocation and policy formulation on the layout of public facilities in urban and rural areas.

1. Introduction

The allocation of medical resources is a critical aspect of urban planning and public health policy, significantly influencing the quality of life in urban and rural areas (Ardeshiri et al., 2018; Barzylovych et al., 2021; Wu et al., 2022). This allocation can follow different approaches, broadly categorized into market-oriented and planning-oriented strategies. Market-oriented allocation relies on the forces of supply and demand, where medical facilities are established based on market dynamics and profitability (Lu & Hsiao, 2003; van de Ven, 1996). In contrast, planning-oriented allocation is driven by governmental policies and regulations to ensure equitable access to medical services across different regions, focusing on equal and dispersed development (Mougeot & Naegelen, 2018).

Real estate prices are a key indicator of economic vitality and living environment quality (Barreca et al., 2020; McDonald, 2012). Several factors influence these prices, in addition to the essential characteristics and conditions of the houses themselves. These include infrastructure, amenities, and public services (Yuan et al., 2020; Zhang et al., 2019). Among these, the availability and diversity of medical resources play a crucial role. Proximity to hospitals, clinics, and specialized medical centers can make a community more attractive to potential homeowners and investors, significantly increasing property values (L. Liu et al., 2022; Peng, 2021). In the study, real estate prices are utilized as a proxy variable for spatial value (Bishop et al., 2020; Leguizamon & Ross, 2012; Wertenbroch & Skiera, 2002), reflecting the willingness to pay for the benefits associated with accessible and diverse medical resources. However, the impact of medical resources on real estate prices shows hierarchical differences (Gu et al., 2024). These differences may also be based on the goals of market-oriented and plan-oriented allocation strategies.

Planning-oriented medical facilities, which aim for balanced and dispersed service distribution, often target rural and underserved areas. The impact of these facilities on real estate prices is either negative or insignificant, reflecting their primary role in serving remote and less developed regions. Conversely, market-oriented facilities tend to cluster in more developed and profitable urban areas, potentially driving up local real estate prices. Understanding the correlation between real estate prices and medical resources is critical. For urban planners, it provides information for resource allocation and infrastructure development, ensuring balanced and equitable urban growth. Furthermore, this knowledge helps policymakers design effective health and urban planning policies to promote economic growth and social equity. It also improves public health outcomes by improving access to care and

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quality of life, making communities more desirable and potentially increasing property values.

Despite the importance of this topic, the comparative impact of market-oriented versus planning-oriented medical resource allocation on real estate prices remains underexplored. This paper aims to fill this gap by investigating how these differing approaches influence real estate prices within the same regions and across different regions. Additionally, it examines the role of diverse medical resources, such as hospitals, clinics, and pharmacies, in shaping real estate values.

The study contributes to the literature in three key areas: 1) Adopting the perspective of overall medical resources, this study systematically analyzes the overall spatial configuration of medical resources at all levels, from grassroots pharmacies to medical centers, filling the gap with previous studies that only focused on specific types or single-story facilities (Gu et al., 2010; Guller & Varol, 2023; Hsieh et al., 2019). This method reveals the spatial configuration and interrelationships of medical resources at all levels; 2) It confirms the significant positive impact of medical resource diversity on spatial value, filling the gap in past literature that only focused on quantity. Diversity emphasizes the complementarity and resilience of resources at different levels within the system; 3) Compares the differences between market-oriented and planned strategies, and examines the impact of different levels of medical resources on spatial values, aiming to clarify the impact of different resource allocation strategies on the environment.

2. Literature review

The allocation of medical resources and its impact on real estate prices has garnered significant attention in urban planning and real estate economics (Tao et al., 2014). Previous research has explored various facets of this relationship, examining how proximity to medical facilities influences property values and the comparative effects of different allocation strategies (Cortés & Iturra, 2019; Peng, 2021; Yuan et al., 2020). This literature review synthesizes existing studies on the market-oriented and planning-oriented allocation of medical resources, their diversity, and the methodologies employed to analyze their impacts on real estate prices.

2.1. Medical resources allocation

Most countries have implemented a tiered healthcare system to effectively utilize medical resources. These medical institutions can generally be categorized into medical centers, hospitals, health centers, clinics, and pharmacies (Jui-fen et al., 2007; Suter et al., 2009). Equity in health care requires that health needs determine these resource allocation and access to health care (Culyer & Wagstaff, 1993; Gu et al., 2010). It encompasses two fundamentally distinct aspects: horizontal equity, which refers to equal treatment for equal needs (Braveman, 2006; Lairson et al., 1995) and represents the maximum coverage area in spatial terms to minimize duplication (Jui-fen et al., 2007). The other is vertical equity, which pertains to different treatments for different needs (Kreng & Yang, 2011; Mooney, 1987), as this study refers to medical diversity. Vertical equity includes the diversity of specialities within a single level (institution), the number of available options, and the richness of accessible medical institutions.

Each level of the medical facility has different responsibilities and roles in patient care (Bazzoli et al., 1999). Higher-level facilities are larger in scale, involving more equipment and professionals (Jin et al., 2019). However, these facilities are shorter in number and more sparsely distributed, which is consistent with the concepts of the long-tail theory and core-periphery theory (Anderson, 2006; Krugman, 1991). These various levels of medical facilities have different establishment goals, resulting in market-oriented or planning-oriented distribution outcomes.

The concepts of market-oriented and planning-oriented medical facility allocation strategies, and their impacts on healthcare provision, are crucial to understand due to their differing effects. The overall geographical distribution of medical resources can be seen as a trade-off between equity and efficiency, aiming to optimize the siting of medical facilities (Culyer & Wagstaff, 1993). Market-oriented medical resources are driven by supply and demand to determine the location and distribution of medical facilities (Mougeot & Naegelen, 2018). Private entities, such as hospitals and clinics, make decisions based on profitability, patient demand, and competitive advantage (Guller & Varol, 2023; Wang et al., 2020). This allocation concept's characteristics include profit motivation, efficiency, and the potential for innovation and welfare improvement driven by competition (Folland et al., 2024). However, it can lead to inequities, with underserved areas often having fewer facilities due to lower profitability, resulting in geographical disparities (Braveman, 2006; Braveman & Gruskin, 2003; Rudavsky & Mehrotra, 2010; Xu et al., 2024).

On the other hand, the planning-oriented approach involves governmental or regulatory agencies strategically determining the placement of medical facilities based on public health needs, equity, and broader social goals (Mougeot & Naegelen, 2018). This approach ensures equitable access to health services, particularly in underserved and rural areas (Weinhold & Gurtner, 2014). Despite its benefits, the planning-oriented approach can face challenges such as inefficiency, slower responsiveness to market changes, and government failure (Folland et al., 2024).

Both approaches are critical because they address different aspects of healthcare delivery. The market-oriented approach drives efficiency and innovation by encouraging competition and responding to patient demand, while the planning-oriented approach ensures that healthcare services are distributed equitably, addressing public health priorities and long-term needs (Mougeot & Naegelen, 2018; van de Ven, 1996). Ideally, market-oriented and planning-oriented approaches should complement each other, creating a more balanced healthcare system. For instance, private facilities can provide high-quality, specialized services in urban areas (Guller & Varol, 2023), while government planning ensures essential healthcare services in rural and underserved regions (Mougeot & Naegelen, 2018). Governments can use regulations and incentives to guide private investments, ensuring market dynamics do not lead to disparities (Mougeot & Naegelen, 2018). Public-private partnerships and collaborative planning can enhance resource utilization and comprehensive healthcare coverage, aligning goals and strategies from both sectors.

However, conflicts can arise when market forces prioritize profit over public health goals, leading to resource allocation tensions, straining resource allocation and exacerbating health inequalities and gaps in quality of life (Mougeot & Naegelen, 2018). Policy implementation can also face challenges, as most governments lack the funds to provide equal access to medical facilities, with private providers potentially resisting regulations that limit their operational freedom. In this complex medical allocation system composed of market-oriented spatial aggregation and planning-oriented evenly spread development strategies, we try to measure and clarify its effect on the value of surrounding real estate.

2.2. Medical resources and spatial value

The relationship between healthcare resources and real estate markets is increasingly critical, particularly as populations age, and is rooted in residents' demand for essential services and their impact on quality of life. According to consumer choice theory and the concept of "voting with one's feet" (Rosen, 1974; Tiebout, 1956), individuals are often willing to pay a premium to reside near amenities that improve quality of life, including healthcare services (Levesque et al., 2013). It is the same as the YIMBY include libraries, schools, hospitals, museums, metro stations, and large parks (Yang & Su, 2011; Zhang et al., 2019), which influence wages, land prices, and housing prices (Schuler, 1974), and favourable amenities encourage the growth of urban populations, regional economies, employment, and creative industries (Li et al., 2016).

When health care has become an essential residency requirement, this highlights the need for communities to have adequate health care resiliency. Resilience depends on the substitutability of resources, that is, the diversity of medical resources, ensuring that multi-level medical infrastructure can adapt to different spatial needs while providing residents with accessible care. As medical equity theory emphasizes (Culyer & Wagstaff, 1993; Kreng & Yang, 2011), a well-distributed network of medical facilities supports horizontal equity, that is, equal access to similar services, while vertical equity is that of meeting specialized medical needs ability (Du et al., 2023; Fleurbaey & Schokkaert, 2011). This kind of diversified medical services can support diverse population groups and provide other alternatives when a certain medical facility collapses. The uniqueness of the location brought by diversity can promote the improvement of spatial value (Fujita, 1989; Wang et al., 2023). This is consistent with hedonic pricing theory, which states that the value of a property is not only determined by its characteristics, but is also affected by surrounding public facilities, including the availability of medical services.

Regarding the use of real estate prices as a proxy for this specific willingness-to-pay (WTP), Bishop et al. (2020) estimate a flexible housing-price function based on sale prices, physical attributes, and location-specific amenities, where its derivative under ideal conditions reflects the amenity's implicit price. Leguizamon and Ross (2012) used housing prices to be variable with the spatial Durbin hedonic price model to provide marginal WTP estimates for housing units and those of their neighbours. In order to understand homeowners'WTP for proximity to public transportation infrastructure, Macfarlane et al. (2015) conducted a study using the market value of homes as the dependent variable and found that greater distance from transit stations negatively impacts market value.

Traditional hedonic pricing models have shown that property values are influenced by various factors, including location, environmental quality, population size, urban improvements, amenities, and disclosure of environmental risks (Aziz et al., 2023; Haurin, 1980; Lisi, 2019; Michael et al., 2000; Rosen, 1974; Sirmans et al., 2005). Housing values are also linked to property characteristics such as ownership, size, age, number of bathrooms and bedrooms, land area, and housing quality (Can, 1992; Carlsen & Leknes, 2021; Wu et al., 2004). Individuals are willing to pay more for homes with better amenities and accessibility, influenced by the surrounding social and natural environment (Cortés & Iturra, 2019; Tiebout, 1956). However, most studies involving traditional hedonic pricing models have not adequately examined the impact of spatial heterogeneity (Helbich et al., 2014; Li et al., 2016).

The allocation of medical resources will directly affect matters of life and death and residents' quality of life, the supply and demand relationship in the real estate market, and the improvement of regional competitiveness. Market orientation emphasizes demand, supply, and profit-driven efficiency, which is consistent with neoclassical economics and spatial competition theory (Marshall, 1890; Montefiori, 2005), in urbanization, core-periphery theory, agglomeration economy and minimum under the principle of differentiation, medical resources are homogeneously aggregated and unevenly distributed across locations. In this case, medical facilities are often more concentrated in urban areas, which leads to greater competition and innovation (Komlósi & Páger, 2015) but also excludes less profitable rural areas (Braveman & Gruskin, 2003). In contrast, planning-oriented strategies based on welfare economics and public goods theory (Arrow, 1978; Mougeot & Naegelen, 2018; Preker et al., 2000) ensure fairness by allocating medical resources to meet public health needs.

However, under different scenarios, the impact of medical facilities on real estate values is not always positive. Some medical institutions may cause nearby residents to have negative emotions about their existence due to their potential “NIMBY effect”, such as traffic congestion, noise, and crowd problems, thereby inhibiting the rise in housing prices (Lee et al., 2009). Previous research has focused mainly on either market-oriented or planning-oriented medical facilities, examining their allocation independently without integrating both approaches to consider their collective impact on property/spatial value. Additionally, few studies have explored how different levels of medical facilities interact in a single framework (Z. Gu et al., 2024; Jin et al., 2019). This gap in the literature needs to be addressed to better understand the broader impact of medical resource allocation on spatial values. The study seeks to bridge this gap by investigating how allocating diverse medical resources—from primary pharmacies to major medical centres—influences property values. By incorporating market-oriented and planning-oriented allocation strategies, the study aims to comprehensively analyze how healthcare diversity shapes spatial value, offering new insights for urban planners and policymakers.

While urban amenities and accessibility play a crucial role in determining residential locations, the socioeconomic conditions of the community and neighbourhood also influence housing prices (Li et al., 2016). Tajani et al. (2017) analyzed the influence of socioeconomic variables such as property taxes, household incomes, and consumption on housing prices in the Apulia region of Italy, finding significant functional relationships between these variables and property prices. The studied found a strong link between higher education, better employment opportunities, increased income, and higher real estate values (Bischoff, 2021; Feng, 2021; Liang & McIntosh, 1998; Luo, 2020)

To address these limitations, advanced spatial analysis techniques have been introduced. Multi-stage spatial data mining (Yuo & Tseng, 2020) and multiscale geographically weighted regression (MGWR) models offer robust frameworks for analyzing spatially distributed data and understanding localized variations in the impact of medical resources on real estate prices. These methods provide a more nuanced understanding of how medical facilities influence spatial values across different scales and contexts.

Multiscale geographically weighted regression (MGWR) models represent a significant advancement in spatial analysis, allowing for the examination of spatially varying relationships at different scales (Fotheringham et al., 2017). MGWR models have been applied in urban studies to assess how the impact of various factors, including amenities, medical resources, on real estate prices changes across different spatial scales (Helbich et al., 2014; Mansour et al., 2021; Wu et al., 2019). These models provide a more detailed and localized understanding of the spatial heterogeneity in real estate markets.

2.3. Study case

Taiwan, located at the Northeast and Southeast Asia intersection, covers approximately 35,873 km^2. Its long, narrow shape is dominated by the Central Mountain Range running north to south. The terrain is mainly mountainous and hilly, accounting for two-thirds of the island's area. With a population of 23 million, Taiwan implemented the National Health Insurance (NHI) program in March 1995, aiming to reduce financial barriers for all residents through a comprehensive, unified, and universal healthcare system (Kreng & Yang, 2011). Taiwan's medical and healthcare system was initially government-provided, with a 7:3 ratio of public to private hospital beds in the 1960s, but as private hospitals gradually entered the market, they became dominant by the 2010s, accounting for 70 % of hospital beds (Lin, 2012). Under this evolving hierarchical structure, medical resources are now divided into six levels, from top to bottom: medical centers, regional hospitals, local hospitals, health centers, clinics, and pharmacies. Different levels of medical facilities exhibit distinct characteristics of market-oriented and planning-oriented strategies, shaped by their establishment purposes, history, operational models, resource allocation mechanisms, and reimbursement structures under the National Health Insurance (NHI) system.

Firstly, medical centers and regional hospitals demonstrate clear market-oriented characteristics. These large-scale medical institutions, having passed hospital evaluation by the Ministry of Health and Welfare, are designated as institutions with multiple functions such as research, teaching, training and advanced medical operations(Ministry of Health and Welfare, 2013), and are predominantly located in economically developed and densely populated urban areas,[1] catering to the demand for specialized, high-quality healthcare services. Their operational models are highly responsive to market demand and competitive pressures, with revenue streams largely dependent on non-medical surpluses (such as parking lots and food courts), medical supplies, testing reagents, and price differences of medicines.

Conversely, public health centers and local hospitals are established in remote or underserved areas to ensure healthcare access and resource equity (Huang & Soong, 2013; Wu et al., 2021), following government planning. Local hospitals serve as crucial links between primary clinics and regional hospitals, providing acute medical care, inpatient services, surgical treatments, as well as chronic and long-term care and preventive health services (Huang & Soong, 2013). Health centers are responsible for community nursing, prevention, vaccination, nutrition, health education, and medical management, as well as providing outpatient services, mobile medical care, and emergency care (Health Promotion Administration, 2021).

Taiwan's market-oriented healthcare system reflects its free-enterprise economy, with a mix of public (35 %) and private hospitals (65 %), where approximately two-thirds of physicians are hospital-employed and the remainder are private practitioners operating on a fee-for-service basis (Lu & Hsiao, 2003). Rural areas typically experience physician shortages, whereas urban areas have a high concentration of doctors, particularly specialists (Kreng & Yang, 2011). Pharmacies and clinics are established by private individuals according to the results of the market economy by market forces such as demand, purchasing power, and private sector investment, thus, a centralized approach to the distribution of medical resources.

Taiwan's healthcare system represents a hybrid model that integrates market-driven and government-planned approaches, balancing accessibility, efficiency, and competitiveness. Similar to the United States and Japan, Taiwan's healthcare institutions are predominantly private (approximately 65 %), fostering a competitive environment that aligns with market demand (Kuo, 2009; Lu & Hsiao, 2003). However, unlike the highly privatized and multiple-payer U.S. system, Taiwan operates under a single-payer NHI scheme(Hsiao et al., 2019), which ensures universal coverage and cost regulation; aligning it more closely with the healthcare models of the United Kingdom, Germany, and Japan. In contrast, the United Kingdom's National Health Service (NHS) adopts a predominantly government-planned model, most hospital care for NHS patients is provided by public hospitals, ensuring equitable access but often at the cost of longer wait times and capacity constraints (Alderwick, 2022; Moscelli et al., 2021). Japan presents a more comparable case, as it combines universal health insurance with predominantly private healthcare institutions (Campbell et al., 2014).

Both countries exhibit spatial differentiation in market- and planning-oriented healthcare institutions. But in Japan's healthcare system, there is no clear distinction between hospitals and private clinics (Ikegami & Campbell, 1999). While national public and social security-owned hospitals provide advanced care and private hospitals focus on primary services, hospital resources remain concentrated in urbanized, high-population prefectures due to market profitability, with local public hospitals—constituting only 11 % of hospitals—serving as key but limited countermeasure by local governments to ensure healthcare access in less developed areas (Zhang & Oyama, 2016), akin to Taiwan's local hospitals and health centers. Large medical centers or hospitals in both countries remain market-driven, primarily situated in metropolitan areas. This spatial organization reflects a balance between government intervention and market dynamics, promoting both equity and efficiency.

The real estate market in Taiwan is diverse, with significant variations between urban and rural areas. Major cities like Taipei, Taichung, and Kaohsiung exhibit high real estate prices driven by economic development, infrastructure, and access to amenities, including healthcare services. In contrast, rural areas show relatively lower property values, influenced by factors such as limited infrastructure and economic opportunities. The allocation of medical resources in Taiwan reflects these urban-rural disparities (Jui-fen et al., 2007).

Specific characteristics of medical resources and real estate prices are highlighted in various studies. Gu et al. (2024) showed that higher-tier hospitals are spatially associated with higher housing values in China. Similarly, in the United States, Li et al. (2016)found that housing values tend to be lower near hospitals, as high-level healthcare facilities are often located in urban centers. Tran et al. (2023) demonstrated that in the context of housing wealth and medical utilization, a 16 % average decrease in housing wealth could lead to a 0.4 % reduction in prescription drug usage, a 0.5 % decrease in outpatient services, and a 0.6 % decline in the number of doctor visits. However, in South Korea, cultural attitudes towards death and mortuaries differ from other countries, resulting in a negative impact on housing prices near hospitals (Huh & Kwak, 1997). Research by Lu et al. (2024) in Thailand suggests that the urban status of Chiang Mai renders the impact of hospitals on housing prices ignored.

Various studies in Taiwan show differing results. Yuo and Yang (2021) found a positive correlation between the diversity of clinics, hospitals, and health stations and real estate prices. Similarly, Peng (2021) reported that the accessibility of spatial healthcare for the elderly positively impacts property prices. However, people tend to avoid living near hospitals due to unpleasant noise, crowds, traffic, and potential health hazards (Lee et al., 2009; Peng & Chiang, 2015). However, Lee et al. (2009) measured the "not-in-my-backyard" (NIMBY) effects for 67 urban facilities in the Taipei metropolitan area. They found that hospitals and educational facilities like schools have mild NIMBY effects due to traffic congestion and noise. Lin and Lin (2014) used the hedonic pricing method to evaluate the impact of the number of medical facilities on housing prices in Taipei, but no significant effect was found. Most of these studies focus on the capital or a single county, leaving the broader impact of different medical resources on property prices across Taiwan less understood.

The study examines the impact of various levels of medical facilities on real estate prices and explores the importance of medical diversity. Through this analysis, we aim to investigate the differential impact of medical richness and diversity on real estate prices. Furthermore, based on empirical results, we analyze the relationship between the allocation of medical resources and property prices across Taiwan, summarizing the impact of different medical resources on nearby property values.

3. Methodology

3.1. Data and assumptions

The focus of this study is to present the correlation between Taiwan's comprehensive medical resources and real estate prices and the adequacy of the spatial coverage of each medical level. Therefore, Taiwan's main island's spatial scope and complete medical level are used as the research scale. The study uses Taiwan's medical classification: medical centre, regional hospital, local hospital, health center, clinic, and Pharmacy. The above six levels are considered Taiwan's medical diversity, and their expected results on property prices are set based on the above-mentioned relevant literature. The study data was obtained in 2020 and collected on the "Hospital Information Public Platform of the Ministry of Health and Welfare". The research variations and parameter

[1] According to Taiwan's hospital evaluation and teaching hospital evaluation operating procedures, one medical center is evaluated for every two million people, and the number of medical centers in the country and in each first-level medical region is estimated.

settings of medical diversity are listed in Table 1.

The "average unit price" variable used in the real estate price is the actual price registration data released by the Ministry of the Interior. It integrates the real estate transaction prices in the third quarter of 2013–2019(A total of 387,564 real estate price sample point), deducts the land and parking spaces, and presents the price gradient of the existing living space in each village. After conducting normality tests on the variables, we adjusted the average unit price to its natural logarithm, denoted as LnAveUnitPrice. (Bischoff, 2021; Can, 1992). In addition, we use neighbourhood characteristics, accessibility, and amenities as controls (Dantzler, 2021; Li et al., 2016; Peng & Chiang, 2015; Yuan et al., 2020), which contains population density variables from social and economic statistics from the Department of the Interior, and accessibility and convenience variables from 14 transportation facilities(mass rapid transit, bus stop, high-speed rail station, intercity bus station, airport terminal, bicycle station, bike path, national highway, dock, highway bus stop, train station, light rail station, pedestrian walkway, car park), and 8 amenities(park, police station, social welfare institution, fire station, sports facilities, library, environmental pollution control facilities, outdoor recreation) using the MSDM methodology is also used to calculate their diversity values(Yuo & Yang, 2021).

The study setting up different buffer (service) ranges for various medical levels, reflecting the distinct resources and services each provides. Compared to local hospitals, which focus on multiple medical departments and critical patients, clinics mainly address single outpatient categories and mild symptoms, resulting in a more enormous scope of supply and service for unique and diverse medical resources. The buffer zone determined by distance, which assumes an optimistic coverage range provided by suppliers, is consistent with all medical supply service literature and will impact the study results (Hsieh et al., 2019). The assumed range will impact the study results. Therefore, it is necessary to refer to previous medical literature for distance designations and consider the differences in spatial scale, transportation, and healthcare-seeking habits between foreign studies and Taiwan. The distance settings (in Table 1) for different levels of medical services in this study are as follows:

Lue (1995) considered the medical treatment time of outpatients in local hospitals for 27.6 min, at a reasonable speed of 30 km per hour in the city centre, and considered past research to assess medical resources and the accessibility of Western physicians (Chang et al., 2011; Hsieh et al., 2019), so the regional hospital is set to 15 km. Compared with regional hospitals, the medical centre provides medical work for teaching, research, and critical care. Therefore, the impact distance is far from the regional hospital, so the medical centre is 20 km. Wu (2008) proposed that regional hospitals' emergency medical treatment distance is 7.5 km, so the area of influence of local hospitals is set to 7.5 km. Based on the literature, the travel time to hospitals is assumed to be around 30 min. Pharmacies, serving more immediate needs, are estimated to be within a 5-min travel range, roughly 1 km. Primary clinics, addressing more significant medical demands, are estimated to be within a 10-min travel range, approximately 3 km. Finally, health centers, which play a crucial role in disease prevention and regional health management, are considered to have a broader service range of 5 km.

Table 1
Variable name and parameter setting.

Code name	Category	Amount of data	Buffer distance	Expected result
M1	Pharmacy	6565	1 km	+
M2	Clinic	21,397	3 km	+
M3	Health center	353	5 km	−
M4	Local hospital	358	7.5 km	−
M5	Regional hospital	82	15 km	+
M6	Medical centre	25	20 km	+

3.2. Research design

To rapidly assess and identify the effects of environmental product diversity and heterogeneity, this study employs the Multi-stage Spatial Data Mining (MSDM) method proposed by Yuo and Tseng (2020). This method simulates the environmental product diversity of medical services using concepts of externalities, product diversity, and spatial distribution. It includes various measurement standards and examines their relationships with urban and rural real estate prices. Compared to previous diversity measures such as the Shannon Entropy index, Simpson Diversity, and Herfindahl-Hirschman index, MSDM offers more direct applications, enabling the visualization of diversity distribution in geographic spaces through 2D or 3D models.

The MSDM methodology is designed based on a series of theories. Firstly, it is grounded in agglomeration economy theory, drawing on scale and scope economies from Marshallian and Jacobian externalities. It studies how the clustering of retail activities generates positive spillover effects, highlighting Meade's (1952) concepts of unpaid factors and atmosphere(i.e. buffer) and Peng and Tabuchi's (2007) optimal variety conditions under monopolistic competition in locational environments. Secondly, the methodology employs the retail hierarchy concept from central place theory (Clark & Rushton, 1970), focusing on discussions of maximum product diversity and different levels of goods, including the classic Hotelling (1929) spatial competition process under the principle of minimum differentiation (Meagher, 2012) and consumer preference ideas. Lastly, it incorporates Krugman (1991) core-periphery model, applying it to the spatial distribution of urban resources with increasing and constant returns to scale.

Based on the external effects generated by the locations of medical resource providers on the environment, Geographic Information Systems (GIS) are used to extract levels of product variety within different service distances. Diversity indices are calculated to simulate the differences in service scales across various categories. Finally, regression analysis and cluster analysis are employed to examine the impact of medical diversity on real estate prices. Therefore, this study can be divided into six stages for the spatial data mining process of product variety (medical institution categories):

Stage 1. Data preparation of the selected areas. In this research, four basic data sets were used: (a) the points of each medical resource/institutions location, (b) property price data points, (c) Village administrative boundaries and (d) the basemap from OpenStreetMap. All the subsequent extension data sets and variables were generated from these four basic data sets, such as the price agglomeration areas. The required tunings of the raw data, e.g. the price is taken as the natural logarithm and medical supply area identification, were also completed at this stage.

Stage 2. Defining and identifying medical diversity. To extract the layers of medical categories, the classification and definition system of product variety was crucial for obtaining research results. The design of index elements relies on the research objectives. This study is the first to extract layers of medical categories for a 3D visual demonstration of spatial distributions. Consequently, a tiered healthcare system was utilized based on the most common differences in medical features (Table 1), encompassing six medical categories. After categorizing all points into these six medical categories, the layers of medical categories were established.

Stage 3. Generating the spatial distribution of the agglomeration areas for each layer of the medical category. This stage involved generating all 6 layers of retail categories separately. Each layer presented the spatial distribution of agglomeration areas of the single product category. The process for determining the agglomeration areas is described as follows: (a) all points of retail supply were classified into 6 tiered medical categories, (b) the layer of each medical category was selected and exported, and (c) each level was buffered according to the effect/supply services range(refer to buffer distance in Table 1).

**Stage 4. Generating village buffer areas: capturing rents and

other variety features. These village buffer areas (VBAs) become the basic measurement units of performance data (i.e. prices). Hence, other than the property values, the number of institutions/levels, the coverage of medical facilities on each village, and the calculation of medical diversity indices were based on these VBAs. Data mining processes such as multiple regression, MGWR, clustering analysis and the 3D price gradient were also based on VBAs of each medical supply area.

Stage 5. Connecting all data sets: generating medical diversity indices.

The major task at Stage 5 was to establish a connection between data sets and generate more diversity indices from it. The process involved (a) using ArcGIS Pro to establish the link among medical supply points, prices and spatial features; and (b) generating diversity indices (e.g. Shannon, Simpson diversity and Herfindahl indices). Fig. 1 shows the combined two-dimensional (2D) visualization of medical diversity indices. Diversity indices include at least three elements: the fragments of various medical buffers from neighbourhood medical agglomerations, the number of layers for each fragment, and the combination of contents (different layers) within each fragment. This ***Diversity*** could be generated using spatial geoprocessing tools and spatial intersects with the measuring grid system. Table 2 shows the results and descriptive statistics for the entire medical area. The number counts (L) for each variety layer (L = 1–6) in Table 1 represent the total number of medical layer types within each fragment. $N = 1 \ldots n$ is the individual resource under each type of layer L.

$$\boldsymbol{Diversity}_{fLi} = \sum_{f}\sum_{L} z_{fi}(x_L) \forall f, L, i \tag{1}$$

This means that for any M_{LN}, the buffer abundance for its location i ($\boldsymbol{Diversity}_{fLk}$), is the sum of fragments of different buffer textures at location i, and the total layer counts within each fragment ($\boldsymbol{f}$). The buffer abundance provides the medical category richness, store abundance or density, and combination variance within the measuring unit. Each fragment represents a certain medical buffer provided by the environment. Medical diversity is provided by the differences in services provided by medical resources at the same level, such as the differences in the specialities of clinics and doctors and other differences caused by uniqueness.

Stage 6. Generating data mining results. After the extraction stages, the final stage involved mining for detailed results by including (a) a regression model for examining the main hypothesis and other product variety features, (b) the TwoStep cluster analysis for VBAs and (c) 3D illustrations. To understand the significant contributions of each layer, this analysis involved using the distribution of the frequencies of the 6 layers in all VBAs to generate potential factors.

The purpose of the regression test of the diversity of medical resources processed in advance and taking the logarithm of the average real estate price to detect the impact of resources on real estate prices; finally, the results of cluster analysis will be used to examine the distribution patterns of Taiwan's medical resources and real estate prices.

3.2.1. Ordinary least squares (OLS)

One of the most famous regression analysis techniques for identifying mechanisms within the real estate market is the ordinary least squares (OLS) method (Isazade et al., 2023). OLS linear regression is mainly used to regress a dependent variable on other explanatory variables. This technique assumes a stationary and constant relationship across space, which may render the implicit independence assumptions related to the geographical data invalid (Hutcheson, 2011; Mansour et al., 2021). This study analyzes the relationship between the diversity of medical resources and the spatial values (Y) by establishing a line of best fit.[2] The OLS model is expressed as follows:

$$Y_i = \beta_0 + \beta_{m1}MD_{1i} + \beta_{m2}MD_{2i} + \beta_{m3}MD_{3i} + \beta_{m4}MD_{4i} + \beta_{m5}MD_{5i} + \beta_{m6}MD_{6i} + u_i \tag{2}$$

where Y_i signifies the dependent variable (LnAveUnitPrice) at the i-th location (village level). The independent variables include the diversity of pharmacies, clinics, health centers, local hospitals, regional hospitals, and medical centres, represented as MD_{1i}, MD_{2i}, MD_{3i}, MD_{4i}, MD_{5i}, and MD_{6i}, respectively, where i represents the village. The error term u_i is the interference factor that cannot be explained by the normal distribution of the 6 variables.

3.2.2. Multiscale geographically weighted regression (MGWR)

Although the OLS model assumes consistent relationships between variables across locations, it does not account for spatial heterogeneity, oversimplifying the underlying spatial processes. In many cases, a fixed spatial scale is invalid as phenomena operate at different scales. MGWR addresses this by allowing the relationship between the response and explanatory variables to vary spatially and at different scales (Fotheringham et al., 2017), providing a more accurate understanding of spatial non-stationarity and improving the analysis of geographically dispersed data. The method is also widely used to correct the shortcomings of OLS in spatial adjustment (Helbich et al., 2014; Isazade et al., 2023; Mansour et al., 2021). It incorporates various bandwidths across the study area and is calculated as follows:

$$y_i = \sum_{j=0}^{m} \beta_{bwj}(u_i, v_i) MD_{Lij} + \varepsilon_i \tag{3}$$

where b_{wj} represents the bandwidth for calibrating the j-th conditional relationship (Fotheringham et al., 2017).

A robust regression process in the model, including White's adjustment and weighted least squares, was used to address heteroscedasticity and ensure consistent standard errors and covariance. Furthermore, we also tested for non-linear relationships using transformations such as logs, square roots, and quadratic functions. The final mining process at this stage involved using a basic clustering algorithm to explore the characteristics among different VBA clusters. SPSS's TwoStep cluster component, a scalable algorithm capable of handling both continuous and categorical variables, was used to identify the specific variety characteristics of each cluster.

4. Empirical analysis of medical diversity and real estate prices

The study area is the main island of Taiwan, as shown in Fig. 1. The black outlines in the figure represent the administrative districts. The digital elevation model (DEM) was incorporated into the figure. The dark green in the figure represents a higher altitude; white is gentle. Fig. 2 illustrates the variety of medical facilities at different levels, indicating the diversification of available medical treatment options within a certain distance. The red area shows the diversity of institutions at each level. The darker the color, the more abundant it is. The lighter the color, or even white, it means lack. This shows that the distribution of medical resources in the center of the research area is mainly affected by the terrain.

For medical, the geographic information system is used for multi-stage data mining and analysis, and the entropy value of diversity is

[2] The study uses real estate prices only as a proxy variable for spatial value. It aims to measure the impact of the diversity of medical resources on spatial value and does not focus on the main factors of real estate increases, as this requires consideration of more economic environment trends and changes in social factors.

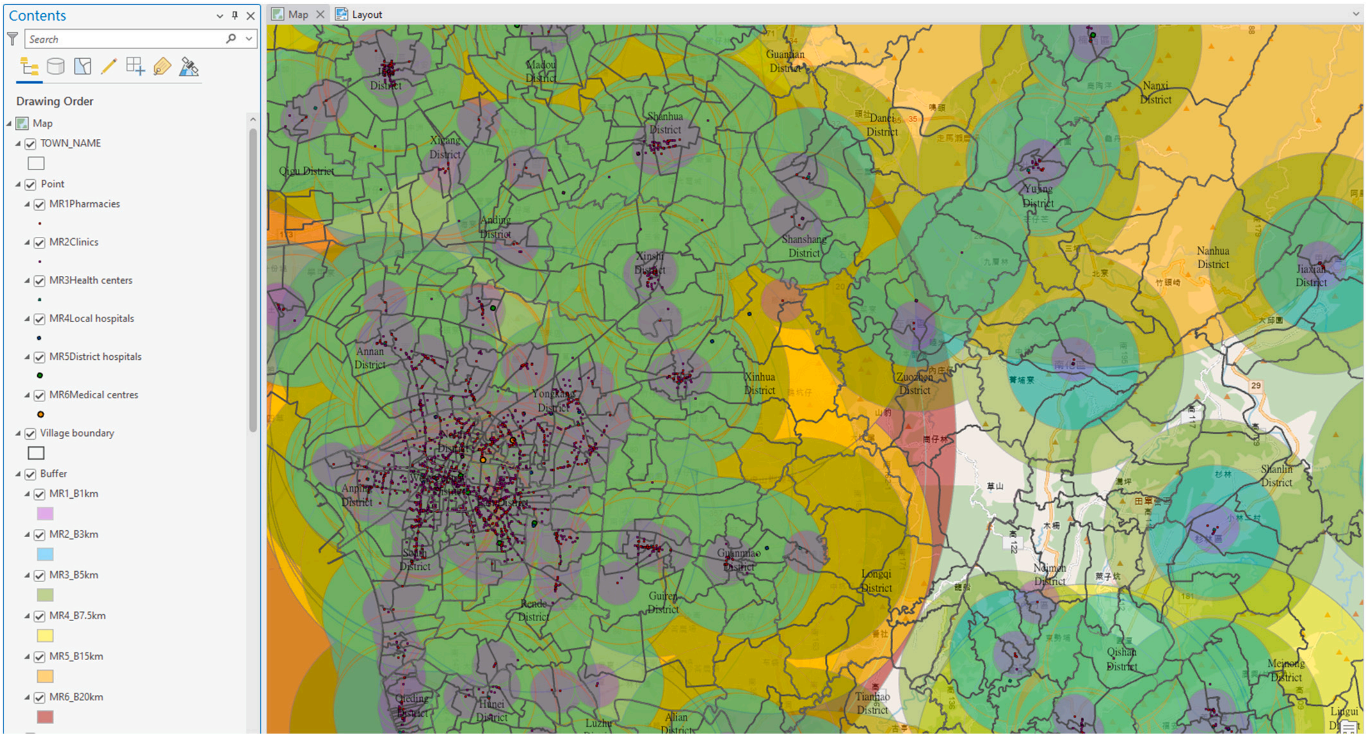

Fig. 1. The process captures features of each medical supply point and VBA.

Table 2
The descriptive statistics of medical diversity.

Category	Min	Max	Mean	SD
Pharmacy Diversity	0	2019	291.7	376.1
Clinic Diversity	0	48	13.1	13.4
Health Center Diversity	0	19	6.1	4.4
Local Hospital Diversity	0	13	5.1	4.9
Regional Hospital Diversity	0	61	5.6	9.3
Medical Centre Diversity	0	9	2.4	2.1
AveUnitPrice	111	505,560	55,814.2	46,822.1
LnAveUnitPrice	4.7	13.1	10.7	0.8

calculated, which is used as an index for analyzing medical diversity. In Fig. 2, pharmacies have very sparse and weak overall coverage despite having the highest single-point diversity at 61, with high concentration and overlap in certain areas. Clinics exhibit the richest diversity among all medical categories. On the other hand, health centers are relatively dispersed and have the lowest regional overlap, with a value of 9. Local and regional hospitals show higher scores in Taipei, Taichung, and Kaohsiung, the three major cities. The real estate prices range from NT $505,560/square meter to NT$111/square meter, and the average rent is NT$55,814/square meter in Taiwan. Descriptive statistics regarding the results are presented in Table 2.

In this study, the village coverage of different medical resource diversity is listed in Table 3. The number of villages lacking medical resources but having real estate transactions shows the current situation of human settlements lacking medical resources. There are a total of 7683 research units. After deducting the areas without real estate transactions, there are 6640 observations (villages).

4.1. OLS regression analysis results

This study uses the OLS regression model, please refer to Table 4a, and 6 independent variables are included for analysis. Table 4a details the regression analysis results of the diversity of different medical levels and real estate prices, showing the significance of multiple variables on the average unit price. The R square of the linear model is 0.626, which means that the model can explain 62.6 % of the real estate unit price. Regarding the empirical analysis, although the value is not high, it is acceptable. The F value test exceeds the 99 % significant level, and the VIF test value among variables is lower than 5, indicating no redundancy among each explanatory variable.

The regression results indicate that most medical resources positively influence real estate prices. Regional hospitals have the most substantial impact, with a coefficient of 0.47, followed by health centres, medical centres, pharmacies, and clinics, with coefficients of 0.0340, 0.0314, 0.0131, and 0.004, respectively. Conversely, local hospitals are associated with a lower coefficient of −0.0082, indicating that local hospitals have a negative correlation or are located in low-price areas.

To more comprehensively reflect the endogenous factors and distinctions between market-oriented and planning-oriented medical resource allocation strategies, we incorporated an interaction term between clinics and local hospitals. Table 5 presents the results of an Ordinary Least Squares (OLS) regression model that incorporates interaction terms to examine the determinants of the logarithm of average price (LnAveP). Importantly, with the interaction term introduced, the previously (Table 4) observed negative effect of local hospitals is now captured within the interaction term, which also has a negative coefficient (−1.7e-05) and is statistically significant ($p < 0.001$), indicating that while clinics alone contribute to price appreciation, their effect diminishes as the number of local hospitals increases. The visualization in Fig. 3 further illustrates the effect of this interaction, showing the relationship between local hospitals and LnAveP under different clinic diversities. The results show a negative relationship between local hospitals and LnAveP across both levels of clinic diversity. However, the downward slope is steeper in areas with higher clinic counts (Q3 = 415.25), indicating a stronger negative effect of local hospitals in these areas.

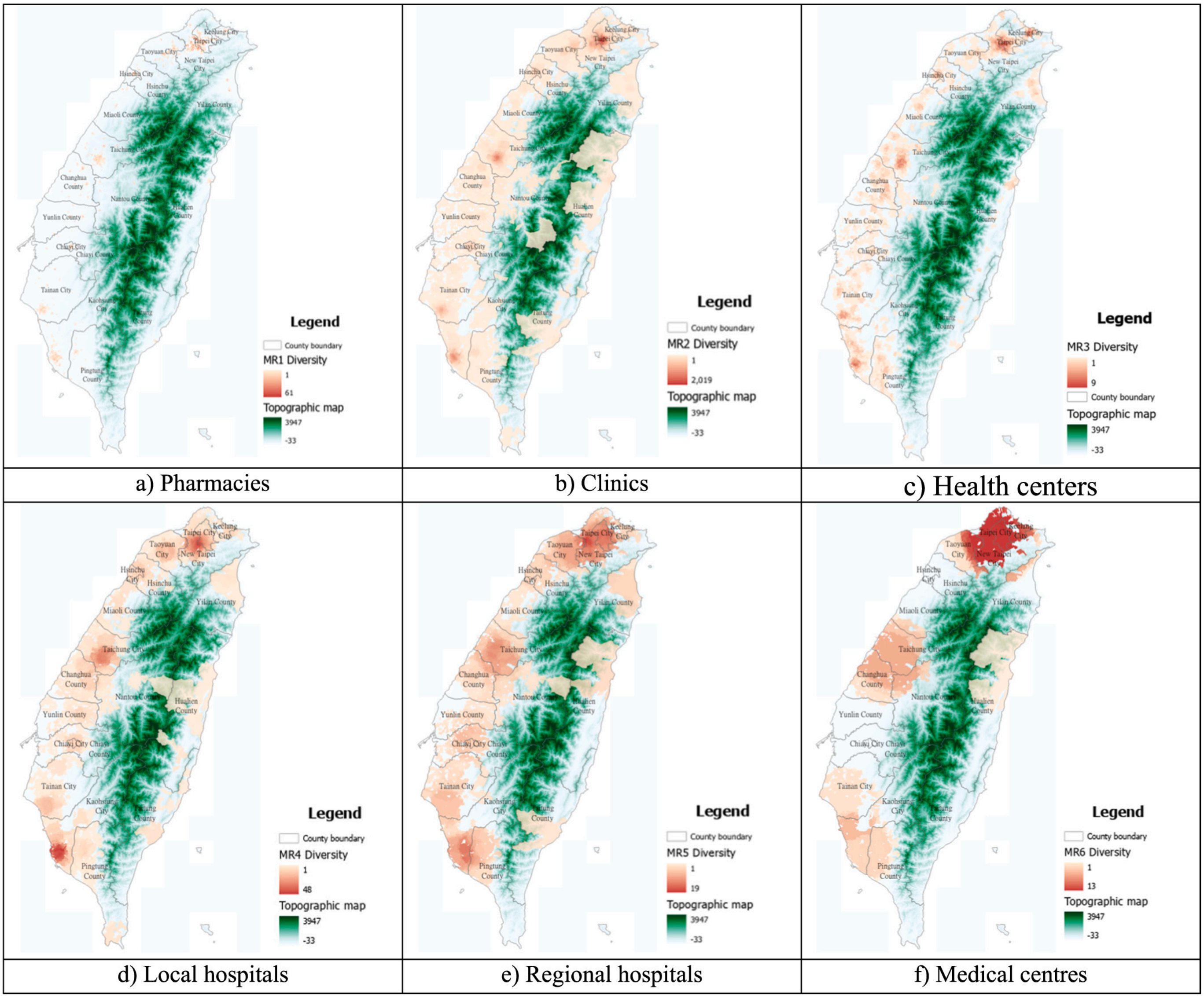

Fig. 2. Diversity distribution of each medical level.

Table 3
Village coverage of diversity of medical resources.

Category	Coverage rate	villages lacking	Villages that lack medical resources but have real estate transactions
Pharmacy Diversity	38.0 %	4762	3767
Clinic Diversity	96.3 %	284	104
Health Center Diversity	78.1 %	1681	1059
Local Hospital Diversity	90.3 %	747	385
Regional Hospital Diversity	91.4 %	658	359
Medical Centre Diversity	70.7 %	2248	1636
The Sum Of All Diversity	98.7 %	97	27

Note: There are 7683 villages in total, 6640 of which have real estate transactions.

4.2. Multiscale geographically weighted regression (MGWR) analysis results

The bandwidth selection of MGWR uses the Golden Search algorithm to determine the number of neighbours for each explanatory variable, iterate between the minimum and maximum number of neighbours, and the final value selected will have the smallest AICc. The MGWR results are shown in Table 4b, which corrects the explanation effect of the traditional OLS model on spatially variable effects. The MGWR and OLS models consistently determine the positive and negative impacts on real estate prices. Most medical resources positively affect property values, with the MGWR mean coefficients revealing a more substantial impact for clinics (0.3639). The other coefficients, from highest to lowest, are Regional Hospitals (0.2344), Medical Centers (0.1750), Health Centers (0.1023), and Pharmacies (0.1007), which closely align with the OLS results. Only Regional Hospitals (−0.0239) exhibit a negative effect. The MGWR model can explain 76.2 % of the variance, which is 13.6 % higher than the linear model.

Fig. 4 illustrates that in MGWR, the real estate price was a significant regressor in describing the spatial distribution of every medical resource diversity in the study area, particularly in the pharmacies, clinics, regional hospitals, and medical centres. The MGWR coefficient map of

Table 4
The results of regression models.

a) OLS Model results

	Coefficient	SE	t-Statistic	VIF
(Constant)	9.8345***	0.0016	615.0905	
Pharmacy	0.0131***	0.0001	17.2993	2.2044
Clinic	0.0004***	0.0000	16.3158	4.5429
Health center	0.0340***	0.0053	6.4234	3.8094
Local hospital	−0.0082***	0.0007	−11.4662	4.3363
Regional hospital	0.0470***	0.0029	16.2568	4.4315
Medical centre	0.0314***	0.0018	17.0677	2.7867
Population Density	1.7704***	0.4246	4.1699	3.0339
Accessibility	0.0022***	0.0002	14.0571	1.7276
Amenities	0.0056***	0.0003	16.2638	1.1788
F	1233.4456			
Sig.	0.0000			
AICc	8712.1438			
R	0.7913			
R-Squared	0.6261			
Adj. R-Squared	0.6256			
Note	a. Dependent Variable: LnAveP			
	b. Number of Observations: 6640 (Only include villages with real estate transactions)			
	c. ***$p < 0.001$.			
	d. White heteroskedasticity-consistent standard errors and covariance.			

b) Multiscale geographically weighted (MGWR) results

Variable	Mean	SD	Minimum	Maximum	Significant (% of Features)	Neighbours (% of Features)
(Intercept)	0.1735	0.3744	−0.9375	1.3610	481 (7.24)	35 (0.53)
Pharmacy	0.1007	0.0019	0.0986	0.1048	6640 (100.00)	6640 (100.00)
Clinics	0.3639	0.0148	0.3456	0.3867	6640 (100.00)	5676 (85.48)
Health center	0.1023	0.1078	−0.0765	0.5135	1944 (29.28)	485 (7.30)
Local hospital	−0.0239	0.0030	−0.0282	−0.0155	0 (0.00)	6640 (100.00)
Regional hospital	0.2344	0.0038	0.2295	0.2429	6640 (100.00)	6640 (100.00)
Medical centre	0.1750	0.0090	0.1653	0.1883	6640 (100.00)	6640 (100.00)
Population Density	0.1188	0.1814	−0.1912	0.7403	1864 (28.07)	419 (6.31)
Accessibility	0.0586	0.0494	0.0030	0.2140	2682 (40.39)	1818 (27.38)
Amenities	0.0654	0.0645	−0.0295	0.2140	2055 (30.95)	626 (9.43)
Sigma-Squared	0.2381					
Sigma-Squared MLE	0.2157					
AICc	10,039.1037					
R-Squared	0.7843					
Adj. R-Squared	0.7619					

Note.
a. Dependent Variable: LnAveUnitPrice.
b. Number of Observations: 6640 (Only include villages with real estate transactions).

Table 5
OLS takes into account interaction terms.

Variable	Coef.	SE	t
(Constant)	9.8125	0.0122	804.7805
Pharmacy	0.0122***	0.0009	13.3722
Clinic	0.0009***	5.7e-05	15.5347
Health center	0.0245***	0.0055	4.4839
Local hospital	−0.0002	0.0012	−0.1260
Regional hospital	0.0373***	0.00261	12.8101
Medical centre	0.0332***	0.0019	17.0992
Population Density	1.8592***	0.5297	3.5102
Accessibility	0.0051***	0.0002	14.4617
Amenities	0.0022***	0.0039	13.1678
Interaction term	−1.7e-05***	2e-06	−9.9613
F	1136		
Sig.	0.000		
AICc	8615		
R-Squared	0.6316		
Adj. R-Squared	0.6310		

Note.
a. Dependent Variable: LnAveP.
b. Number of Observations: 6640 (Only include villages with real estate transactions).
c. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.
d. The interaction term is clinic multiplied by local hospital.

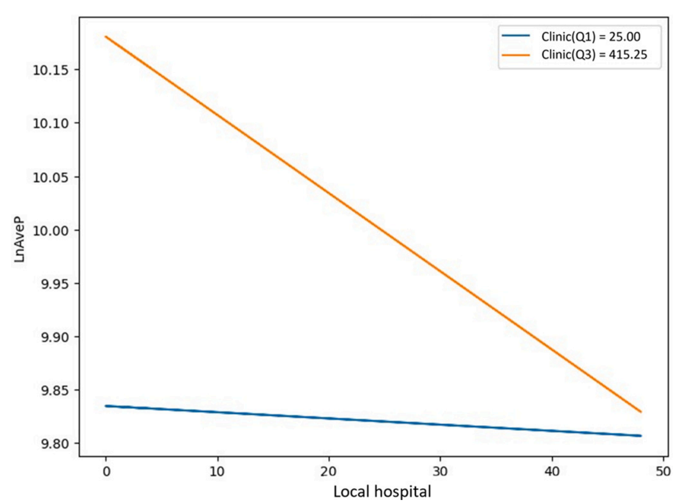

Fig. 3. Impact effect of interaction terms on prices.

real estate price demonstrates that positive estimates in the spatial pattern stretched from most medical resources, except for Health centers and Local hospitals, suggesting a large correlation between property price and medical diversity in these areas. On the contrary, Health

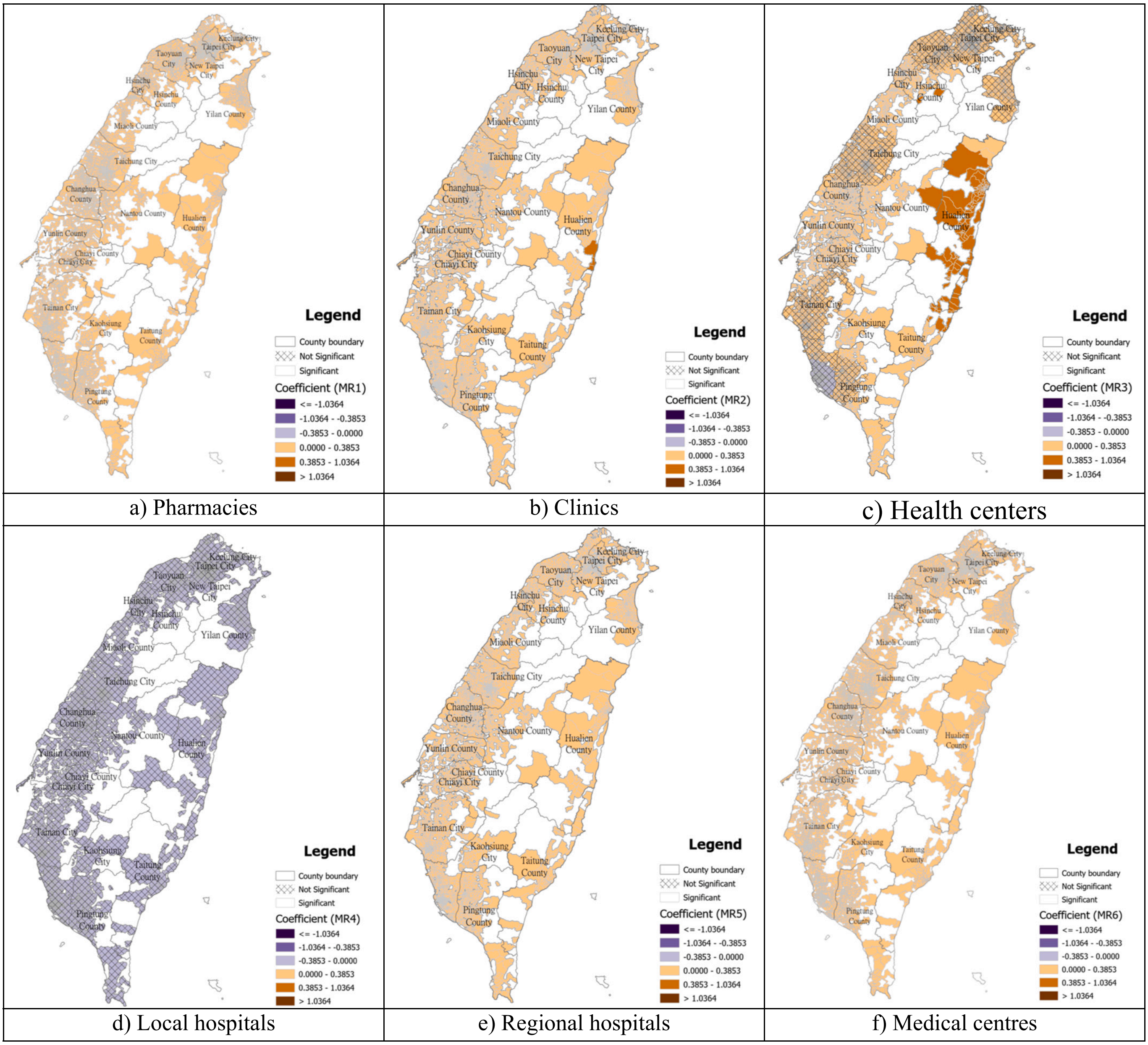

Fig. 4. MGWR coefficient distribution of each medical level.

centers and local hospitals have a negative impact on property prices in some areas, particularly in local hospitals in the north and south. Overall, the larger positive coefficient estimates were found in Taiwan, indicating a stronger relationship between real estate prices and medical diversity.

4.3. SPSS TwoStep cluster analysis results and three-dimensional visualization

Imported Taiwan's medical diversity data analyzed by the geographic information system into SPSS. The input variables include the diversity of medical levels and the average real estate. Unit price, divide the system settings into three clusters, and the obtained analysis results are shown in Fig. 5. The overall cluster analysis quality (Fig. 3a) is fair in the cluster quality measured according to the Schwartz Bayesian criterion. Among the three clusters, the largest cluster is cluster 2, accounting for 47.4 %, with 3145 villages. Real estate prices are the lowest among all clusters, and the diversity of medical resources is also the least. Cluster 1 accounts for 30.9 % and has a total of 2052 villages. This agglomeration has moderate real estate transaction prices and a diversity of medical resources. In Cluster 3, which accounts for 21.7 %, 1443 villages are the areas with the highest real estate transaction prices and also have the greatest diversity of medical resources.

From the relative distribution diagram of each variable value, we can understand that the distribution of real estate transaction prices and medical resources in the Taiwanese villages in Fig. 5b is highly unequal. Areas with lower transaction prices and lower diversity of medical resources make up the majority of rural Taiwan. From the perspective of spatial distribution (Fig. 5c), the diversity of medical resources is highly concentrated in three major urban areas, namely Taipei, Taichung, and Kaohsiung City. Secondly (cluster 1) is distributed around cities or secondary cities. The 3D visualization in Fig. 5d can clearly show the allocated quantity within each medical grade space in Taiwan, and areas with high prices and quantities are presented in 3D and are relatively high on the map.

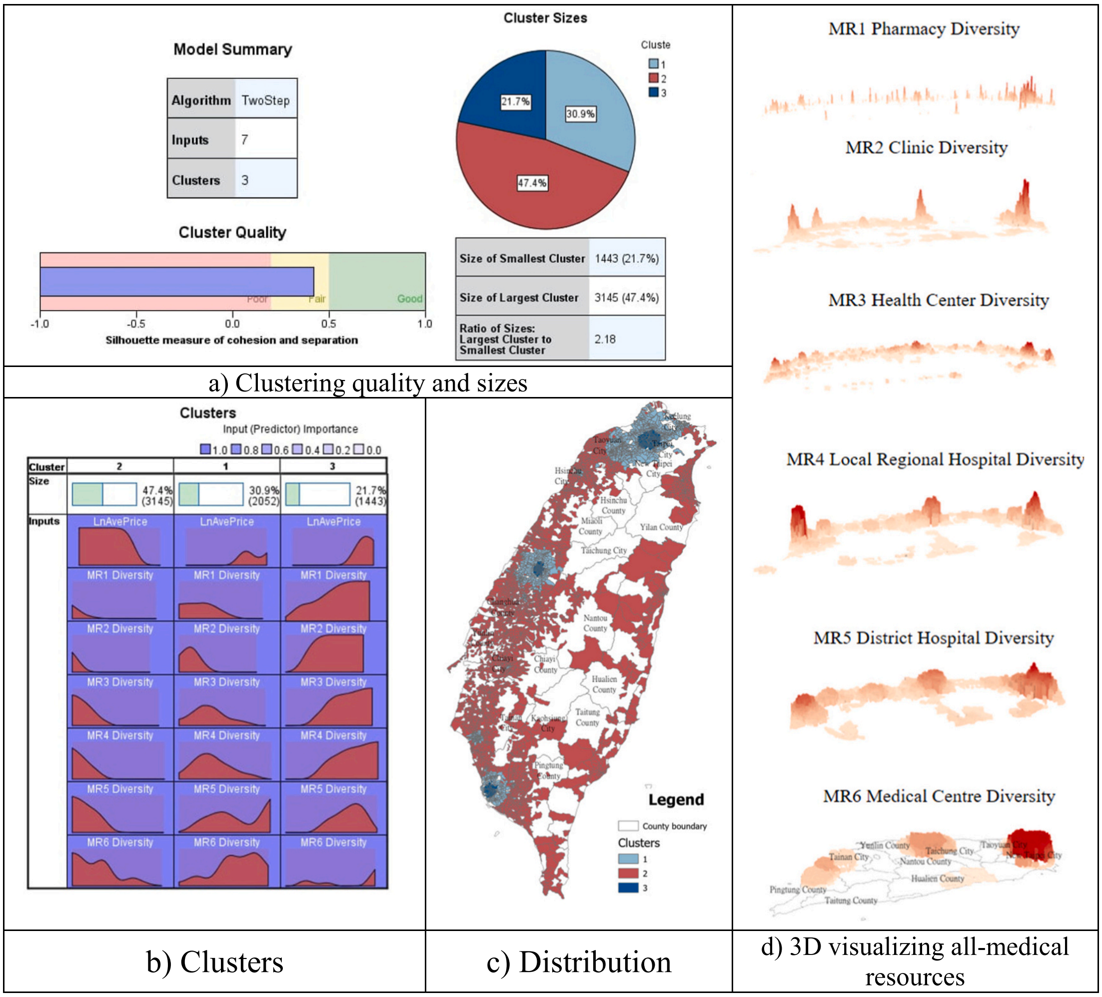

Fig. 5. SPSS Two-Step Cluster results and visualizing medical resources.

5. Discussion

5.1. Medical diversity and real estate price

It is essential to examine the diversity and effective utilization of resources. Byrne et al. (1994) have proposed that sustainable cities should ensure balanced development, pay attention to the diversity of resources, and emphasize efficient resource utilization. Especially medical resources for basic living needs, including the significant impact of the real estate market and regional development. From the analysis of diversity results (Fig. 2 and Fig. 5d), we can observe the spatial distribution and coverage of resources in the region. According to the amount of diversity, check whether there are gaps, duplication or scattered allocation of resources, and check whether there are single or different levels of medical resources between regions (Yuo & Tseng, 2020).

Based on the core hypothesis of this study, an increase in medical diversity leads to greater access to healthcare services. This improvement in quality of life enhances people's preferences, subsequently driving up the real estate price (Levesque et al., 2013). In the cluster analysis, the hypothesis of this study was confirmed. The six levels of medical diversity are all located in areas with higher real estate prices. As the diversity decreases, property prices gradually decrease. This result can be inferred that when there are more medical resources, the diversity is higher, which means that when there is a gap in the allocation of medical resources, other medical resources or other levels can make up for it. This reflects the advent of a resilience concept in the healthcare system (Chaturvedi & Siwan, 2020; Kou et al., 2024). When a medical resource collapses due to a disaster or other impact, other medical resources can replace it. If the diversity is high, the possibility of such a replacement will be higher.

5.2. Market-oriented allocation of medical resource

The primary hypothesis posits that the majority of medical resources, particularly clinics, are expected to meet the population's demands through market-oriented approaches. These approaches tend to concentrate in densely populated areas, while planning-oriented approaches are more evenly spread. Consequently, real estate prices in densely populated areas are naturally higher (Zhou et al., 2020), while those in planning-oriented areas may be lower or negatively correlated with major regions.

The MGWR results indicate that most medical resources positively impact real estate prices. The more market-oriented (concentrated) the resource allocation, the stronger the impact on real estate prices, in the following order: clinics (0.3639), regional hospitals (0.2344), medical centres (0.1750), pharmacies (0.1023), and health centres (0.1007). Market-oriented approaches (such as pharmacies and clinics) prioritize profitability, leading healthcare providers to establish in areas with higher population density and income levels, ensuring a steady customer flow. Research indicates that retail clinics, including pharmacies, are more likely to be found in urban areas with higher median incomes and better-educated populations, as these locations provide a more financially viable customer base (Rudavsky & Mehrotra, 2010). Likewise, it can be reflected in surrounding property values (Yuo & Yang, 2021).

Additionally, the deregulation of community pharmacies has led to an increased number of pharmacies in urban areas, further emphasizing the market's tendency to cluster in regions with higher demand (Vogler et al., 2014).

Contrary to expectations that health centers are typically situated in rural areas where real estate prices are lower (Regan et al., 2003), there is a notable positive correlation between health centers and real estate prices in this study. However, only a small percentage of villages demonstrate significant positive effects. Particularly, there is a significant positive impact in the eastern Taiwan, whereas the effect is not significant in the densely populated metropolitan areas of the west. This means that even in an evenly spread scenario, health centers are still established in desirable residential areas where real estate prices are relatively higher.

5.3. Planning-oriented allocation of medical resource

Policy planning must strategically address the disparities in medical coverage by ensuring that health centers and local hospitals are positioned to fill these gaps. However, this study reveals that planning-oriented allocation remains relatively weak. Local hospitals exhibit a much lower and slightly negative coefficient (−0.0239), suggesting local hospitals in Taiwan are predominantly established through planning-oriented strategies aimed at balancing resource distribution. As a result, they are typically located in suburban or less developed areas rather than in high-priced urban centers, contributing to their negative correlation with real estate prices. This characteristic reinforces their role in supplementing healthcare services in underserved regions. To further substantiate these findings, we introduced interaction terms in the regression model to capture the joint effects of these two explanations. The results show a negative relationship between local hospitals and LnAveP, with a stronger effect in areas with more clinics (Q3).

In theory, medical resources under strong planning-oriented strategies should exhibit a more pronounced negative effect on real estate prices, reflecting a deliberate spatial allocation that prioritizes equitable access over proximity to major urban or high-value areas. However, our study found that only local hospitals showed a negative effect, while basic medical facilities (clinics and health centers) that ideally followed a balanced distribution strategy in the literature still showed a significant positive effect, indicating that there is a mismatch between the target expectations and actual results of primary health care resource allocation (Kreng & Yang, 2011). This study argues that medical resources providing basic medical services and long-term care should emphasize spatially dispersed allocation to offer services closer to the populace, reduce transportation costs, and mitigate the risks of large-scale regional disasters and diseases (Porter, 2010; Simoens & Scott, 2005). However, the weakness in planning-oriented allocation indirectly reflects a potential insufficiency in dispersed allocation.

According to the cluster's result, Table 3, which shows the coverage rates of medical resources, 97 (1.3 %) populated villages lack any form of medical resource diversity, while 27 (0.004 %) villages in areas with real estate transactions—regions where people are most likely to be active—also lack diversity. This is perhaps an optimistic result, as the study assumes that specific medical resources are accessible within a certain distance and does not yet consider population demand or the number of healthcare professionals.

Planning-oriented medical facilities aim to ensure equitable access to healthcare, particularly in rural and underserved areas (Mougeot & Naegelen, 2018). Future policies should target these villages. In addition to medical facility recommendations, planning-oriented resources can be supplemented by telemedicine (Haimi, 2023), drone delivery of essential medical resources (Euchi, 2021), preventive health education, and community centers (Knopf et al., 2016). These services can improve the overall quality of life in rural areas, making them more attractive and positively influencing real estate prices.

6. Conclusions on the characteristics of medical resource allocation and property prices

This study illustrates the importance of the diversity of medical resources and the relationship between their market-oriented or planning-oriented spatial distribution on real estate prices. In the empirical research, the application of the multi-stage spatial data mining method in medical resources was practiced, and the difference in the impact of the diversity of medical resources on real estate prices in market-oriented or planning-oriented development was explained. Most medical resources positively affect real estate prices at regression analysis, even those generally perceived as planning-oriented, such as health centers located in areas with higher property values. The market-oriented allocation of medical resources predominantly influences the distribution of medical facilities in Taiwan, highlighting the current concentration of medical diversity in densely populated areas or urban centers. The cluster analysis shows that Areas with higher medical diversity correspond to higher real estate prices, indicating that the richness of medical alternatives contributes to perceived medical resilience.

However, among the various levels of medical resources, only local hospitals exhibit characteristics of planning-oriented types (detailed in OLS interactions and MGWR for details). A balanced distribution principle is adopted to address issues in rural or underserved areas. Consequently, the spatial distribution of local hospitals shows a negative correlation with real estate prices. When the medical allocation is closer to the market orientation, the impact on real estate prices is more positive and robust, implying medical resources' contribution to enhancing the region's attractiveness and development. Future policy decisions should consider areas with populations but no primary healthcare coverage and the importance of linkages between different healthcare resources.

7. Limitation

Since this study aims to discuss the results of spatial allocation of overall medical resources on a large scale, it has not considered information such as service quality, hospital beds, departments, and the number of doctors in medical institutions. It is suggested that future research can be further included in the discussion. In addition, relevant analysis of demographics and socioeconomic conditions can also be added to expand the scope of research discussions. Healthcare facilities' siting, richness, and impact on real estate prices are complex topics. More research is needed to build evidence using validated measurement tools to assess the impact of medical diversity more collaboratively and analytically.

CRediT authorship contribution statement

Yu-An Yang: Writing – original draft, Visualization, Software, Methodology. **Bo-Cheng Lin:** Writing – review & editing, Validation, Methodology, Conceptualization. **Tony Shun-Te Yuo:** Methodology, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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